

Fuel Rail



Material	
Item	Specification
Motorcraft® SAE 5W-20 Premium Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-20 Super Premium Motor Oil (Canada) XO-5W20-QSP (US); CXO-5W20-LSP12 (Canada)	WSS-M2C945-A

Removal

WARNING: Do not smoke, carry lighted tobacco or have an open flame of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.

WARNING: Before working on or disconnecting any of the fuel tubes or fuel system components, relieve the fuel system pressure to prevent accidental spraying of fuel. Fuel in the fuel system remains under high pressure, even when the engine is not running. Failure to follow this instruction may result in serious personal injury.

WARNING: Do not carry personal electronic devices such as cell phones, pagers or audio equipment of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.

WARNING: Always disconnect the battery ground cable at the battery when working on an evaporative emission (EVAP) system or fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.

WARNING: Clean all fuel residue from the engine compartment. If not removed, fuel residue may ignite when the engine is returned to operation. Failure to follow this instruction may result in serious personal injury.

1. Release the fuel system pressure. For additional information, refer to Section 310-00.
2. Disconnect the battery ground cable. For additional information, refer to Section 414-01.
3. Remove the upper intake manifold. For additional information, refer to Section 303-01B.
4. Disconnect the fuel tube-to-fuel rail quick connect coupling. For additional information, refer to Section 310-00.
5. Remove the LH and RH fuel rail noise insulator shields.
6. Disconnect the 6 fuel injector electrical connectors.
7. Remove the 4 fuel rail bolts.
8. Remove the fuel rail and injectors as an assembly.
9. Remove the 6 fuel injector clips and the 6 fuel injectors.
 - Remove and discard the 12 fuel injector O-ring seals.

Installation

1. **NOTICE:** Use O-ring seals that are made of special fuel-resistant material. The use of ordinary O-rings may cause the fuel system to leak. Do not reuse the O-ring seals.

NOTE: *The upper and lower fuel injector O-ring seals are similar in appearance, but are not interchangeable.*

Install the new O-ring seals onto the fuel injectors and lubricate them with clean engine oil.

2. Install the 6 fuel injectors and the 6 fuel injector clips into the fuel rail.
3. Install the fuel rail and fuel injectors as an assembly.
4. Install the 4 fuel rail bolts.
 - Tighten to 10 Nm (89 lb-in).
5. Connect the 6 fuel injector electrical connectors.
6. Install the LH and RH fuel rail noise insulator shields.
7. Connect the fuel tube-to-fuel rail quick connect coupling. For additional information, refer to Section 310-00.
8. Install the upper intake manifold. For additional information, refer to Section 303-01B.
9. Connect the battery ground cable. For additional information, refer to Section 414-01.